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Task 1: PEAS Specification

System: A Fraud Detection System for digital payments and banking transactions.

Performance Measure: We can measure how successful the agent is through:

- **Precision and Recall:** The percentage of actual fraudulent transactions successfully blocked (Recall) versus the accuracy of its flags to ensure legitimate users aren't incorrectly blocked (Precision).
- **False Positive Rate:** The frequency of triggering unnecessary security checks (like 2FA or account locks) on normal customer behavior.
- **Processing Latency:** The time it takes to approve or decline a transaction (must be milliseconds to avoid payment delays).
- **Financial Impact:** The total monetary value of fraud prevented.

Environment: The agent operates within digital banking infrastructure, payment gateways, and financial networks. The key outside factors it deals with are massive, real-time transaction volumes, fluctuating global time zones, diverse user devices (mobile, web), and actively malicious actors attempting to bypass security.

Actuators:

- Transaction processors to approve, block, or hold a payment.
- Authentication APIs to trigger Step-Up Authentication (e.g., sending an SMS OTP or push notification).
- Alert systems to notify human fraud analysts and generate support tickets.
- Account management protocols to temporarily freeze compromised accounts.

Sensors:

- Data streams from payment APIs (providing transaction amount, timestamp, currency, and merchant category).
- Device fingerprinting tools (reading IP addresses, geolocation, device IDs, and browser types).
- Database queries to pull up the user's historical spending habits and typical locations.
- Threat intelligence feeds listing known malicious IP addresses or compromised card numbers.

Task 2: Environment Classification

- **Fully Observable vs. Partially Observable: Partially Observable.** The AI only sees the digital footprints of a transaction (device, IP, amount). It cannot see the physical real-world context, such as whether the actual account owner is making the purchase or if their phone was stolen while unlocked.
- **Single-agent vs. Multi-agent: Multi-agent.** The environment is highly adversarial. The AI is actively competing against other agents—fraudsters, hackers, and malicious bots—who are constantly adapting their strategies to outsmart the system.
- **Deterministic vs. Stochastic: Stochastic.** Human spending behavior is inherently unpredictable (e.g., sudden travel, impulse purchases), and new types of fraud attacks emerge randomly. The outcome of a single transaction is never guaranteed.
- **Episodic vs. Sequential: Sequential.** The system's decisions affect future states. If the system blocks a user's legitimate card today, they might stop using that card entirely tomorrow. Additionally, fraud often happens in sequences (e.g., a small \$1 test charge followed by a massive \$1,000 charge).
- **Static vs. Dynamic: Dynamic.** The environment changes constantly. Thousands of new transactions occur globally every second, and fraudsters update their tactics while the system is processing data.
- **Discrete vs. Continuous: Discrete.** Unlike a continuous physical process like steering a car, the fraud system evaluates distinct, individual data packets (transactions) and makes a binary, turn-based decision: "Approve" or "Decline."

Task 3: Critical Analysis

1. Greatest Technical Challenge

In my opinion, the biggest technical challenge for an AI engineer building this system is managing the adversarial (Multi-agent) environment combined with an extreme class imbalance. Fraudulent transactions make up a tiny fraction of total global payments. Because the system is Partially Observable, it must constantly guess intent based on limited data. Fraudsters continuously learn the AI's rules and change their tactics (e.g., mimicking normal user behavior or using residential proxies), meaning the AI must adapt in real-time without accidentally locking out thousands of innocent customers.

2. Utility Function and Trade-offs

To manage the trade-off between maximizing security (catching fraud) and minimizing user friction (false positives), I would propose this utility function:

$$U = 0.7 \times \text{Fraud_Caught} - 0.2 \times \text{False_Positives} - 0.1 \times \text{Processing_Latency}$$

If we increased the weight of False_Positives (e.g., making it 0.6), the system would prioritize user convenience over strict security. The AI would become very lenient, rarely blocking transactions unless it was 100% certain it was fraud. This would make the customer experience perfectly seamless and eliminate annoying 2FA prompts, but the trade-off is that sophisticated fraud would easily slip through, causing massive financial losses to the bank.

Structured Representation

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{
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  "peas": {
    "performance_measures": [
      "fraud_catch_rate",
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      "processing_latency_ms",
      "financial_savings"
    ],
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    "actuators": [
      "transaction_approval_blocker",
      "step_up_authentication_trigger",
      "account_freeze_protocol",
      "analyst_alert_generator"
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      "user_history_database",
      "threat_intelligence_feed"
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  "environment_classification": {
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    "agents": {
      "choice": "Multi-agent",
      "justification": "The system actively competes against adversarial agents..."
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    "episodes": {
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      "justification": "Past transactions influence the risk profile, and small..."
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    "dynamics": {
      "choice": "Dynamic",
      "justification": "Thousands of transactions process simultaneously..."
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    "continuity": {
      "choice": "Discrete",
      "justification": "Transactions are evaluated as distinct, separate events..."
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}
```

```
"utility_function": "U = 0.7 * Fraud_Caught - 0.2 * False_Positives - 0.1 *  
Processing_Latency"  
}
```